

# Regression Analysis (II)

## Solutions to the Final Exam.

Dec. 13, 2021

1.(50) Derive the natural (canonical) link function for the binomial distribution in the GLM (generalized linear model).

2.(50) Derive the deviance residual for the Poisson distribution in the GLM.

3.(100) Consider a oneway ANOVA model  $Y_{jk} = \mu + \alpha_j + \epsilon_{jk}$ ,  $j = 1, 2, 3$ ;  $k = 1, 2, 3$ , and the corresponding dataset is given as follows;

trt A 4, 6, 5; trt B 5, 4, 4; trt C 6, 5, 6

(1)(50) When we write the model in matrix form  $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$ , express  $\mathbf{X}$ ,  $\boldsymbol{\beta}$  and  $\mathbf{y}$ .

(2)(50) Show that  $\mathbf{X}^t\mathbf{X}$  is singular, and compute  $\boldsymbol{\beta}$  using the corner point restriction.

4.(50) Let  $f \in L_2[a, b]$  and the sequence of functions  $\{g_j(x)\}$  be CONS for  $L_2[a, b]$ . In writing  $f(x) = \sum_{j=1}^{\infty} \theta_j g_j(x)$ , find the estimate of  $\theta_j$  in terms of CONS and the response variable  $Y_i$ ,  $i = 1, \dots, n$ .

5.(50) Data set  $(X_i, Y_i), i = 1, \dots, 20$  are given as  $X_i = i - 10$ ,  $Y_i = |i - 10|$ . Find the kernel estimator at  $x = 0.5$  using the triangular kernel with  $\lambda = 2$ .

6.(50) Consider a nonparametric regression model  $Y_i = f(X_i) + \epsilon_i$ , and we are interested in minimizing

$$\sum_{i=1}^n \{Y_i - \beta_0 - \beta_1(X_i - x) - \dots - \beta_p(X_i - x)^p\}^2 K\left(\frac{X_i - x}{\lambda}\right)$$

with respect to  $(\beta_0, \beta_1, \dots, \beta_p)$ . Express the above equation in a matrix notation and find the estimator of  $\beta = (\beta_0, \beta_1, \dots, \beta_p)$ .

7.(50) Let  $T_1, \dots, T_n$  be iid survival times with pdf  $f_\theta(t)$  and cdf  $F_\theta(t)$ . Also, let  $C_1, \dots, C_n$  be iid corresponding censoring times with pdf  $g_\gamma(t)$  and cdf  $G_\gamma(t)$ . Assume that  $T_i$ 's and  $C_i$ 's are independent. We observe  $(Y_i, \delta_i)$ ,  $i = 1, \dots, n$ , where  $Y_i = \min(T_i, C_i)$  and  $\delta_i = I(T_i \leq C_i)$  is a censoring indicator under randomly right censoring. Derive a likelihood function for  $\theta$  and  $\gamma$ .